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Vibrational resonance in a bistable van der Pol–Mathieu–Duffing oscillator

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ABSTRACT

In this paper, we study the nonlinear response of a bistable van der Pol–Mathieu–Duffing (VMD) oscillator under the influence of two periodic excitations of widely different frequencies. We have shown that by systematically modulating the strength of the high-frequency drive as well as the strength of the parametric oscillation, a symmetrically oscillating bistable potential can be converted to a symmetrically oscillating monostable potential. In addition to this effect, the strength of the fast drive modifies the damping as well, allowing us to define a threshold value of this strength at which a supercritical Hopf bifurcation occurs. All analytical results have shown to be numerically consistent.

1. Introduction

The response of specific nonlinear systems to a weak signal enhanced by the presence of fast harmonic signal drive has been the central motivation behind all studies concerning vibrational resonance (VR) for the last two decades. After originally proposed by McClintock and Landa [1], a profusion of investigations have been carried out theoretically [2,3], numerically [4] and experimentally [5,6] with the special prominence given to nonlinear potentials [7–16] in the backdrop of VR. Parametrically excited oscillators belong to such a family of nonlinear oscillators, which have been used extensively for the realization of various physical systems in science and engineering [17–22]. The response of a parametric oscillator, whether stable or unstable under the action of biharmonic forces, requires special attention. Although extensive research has been pursued on the response of a high-frequency excited parametric oscillator [23–28], studies of VR with slowly excited parametric oscillators have not, to the best of our knowledge, prominently come to focus [29,30]. We further note that in addition to the time dependence of the natural frequency of a parametric oscillator, if the system operates in a spatially nonlinear dissipative medium, then the response of the system to an additive combination of external low and high-frequency drives is an interesting problem in its own right.

In this paper, we consider van der Pol–Mathieu–Duffing(VMD) oscillator, a typical parametric oscillator with nonlinear damping and two externally acting forcing terms with two widely different frequencies. The intrinsic property of self-excitation of a van der Pol oscillator combines with the parametric excitation allow us to study a more general class of parametric oscillators. Applications of this model can be found directly in various MEMS devices [31,32]. A thorough investigation of non-trivial effects of the high-frequency drive affecting the stability of the slow dynamics through modification of some governing parameters have been considered in recent times [33–36]. However, the nonlinear response of VMD under multiple external forcing with different orders of strength and frequency has remained less explored and is one of our motivation in this paper. In Section 2, the framework of the model and the justifications are described. In Sections 3–5, the detailed mathematical techniques followed by numerical simulations are given. Finally, this article is concluded in Section 6.

2. The model

The typical mathematical model of a VMD oscillator consists of a nonlinear damping term, an oscillatory stiffness and a nonlinear stiffness controlled by two external forcing drives with widely different frequencies. The rationale behind such an arrangement of terms in the context of VR has discussed in the Introduction. The dynamical equation is given by

$$\ddot{x} + \gamma(x^2 - 1)\dot{x} + \omega_0^2(1 + h\cos\omega_p t)x + \alpha x^3 = c\cos\omega t + g\cos\Omega t \qquad (2.1)$$

where $h\cos\omega_p t$ is the parametric excitation with strength h and $c\cos\omega t$ and $g\cos\Omega t$ are the slow and fast frequency drive respectively with $\Omega \gg \omega$. As we consider the parametric excitation to be small, with out loss of generality we can set $\omega_p = 2\omega$ and the reason behind this particular chosen value is clarified in the following paragraph. Now depending on the sign of ω_0^2 and α, three potential structures may arise viz. (a) single well ($\omega_0^2 > 0$, $\alpha > 0$) (b) double well ($\omega_0^2 < 0$, $\alpha > 0$) and (c) double hump ($\omega_0^2 > 0$, $\alpha < 0$). Our model here describes the

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dynamics of a VDM oscillator under a time dependent double well potential,

$$\ddot{x} + \gamma(x^2 - 1)\dot{x} - \omega_0^2(1 + h\cos\omega_p t)x + \alpha x^3 = c\cos\omega t + g\cos\Omega t \tag{2.2}$$

Therefore the oscillator is on the crest of a double well and is in an unstable equilibrium provided the excitation strength h is confined to the domain $-1 < h < 1$. The effect of the high-frequency drive ($g\cos\omega t$) is to redress the double well potential into a single well,thus effectively changing the stability concerning the slow-motion of the oscillator [14,29]. In this paper, we shall show that this high-forcing drive changes the potential structure through an effective modification of the natural stiffness and reshapes the damping function reflected in the slow dynamics of the original system (2.2). Our theoretical study regarding the combined role of the excitation forcing h and the excitation frequencies ω_p and Ω in delicately amending the oscillation frequency and the damping term satisfactorily complies with the numerical results. In order to notice the resonant response to the slow drive $c\cos\omega t$, the contribution of h is only perceptible for non zero fast forcing strength, i.e. $g \neq 0$ and show up in the flow equations which has been elucidated in the next section. Before proceeding to the next section, it is noteworthy that ω_p and ω are entirely two different frequencies and have a subtle role in modifying the system parameters. However, in our paper, we consider $\omega_p = 2\omega$, keeping in mind that our sole motivation is to study the system's response close to the unperturbed or the natural frequency (ω_0) of the system [30]. In our analytical description for deriving the response curve that finally leads to the formation of a limit cycle, we have used multiple time-scale analysis to the first order in perturbation. The choice of the excitation frequency as being twice the external forcing frequency allows the effect of the parametric excitation (h) to be felt in the flow equations at the very first order in perturbation [37].

3. The effective slow dynamics

To analyse the effect of fast forcing drive to the system (2.2), we use the method of direct partition of motion [3]. To implement this method in the system one considers two time scales, one slow and the other fast, so that the dynamical variable can be written as a sum of a slow variable $s \equiv s(t, \omega t)$ and a fast variable $f \equiv f(t, \Omega t)$, viz.,

$$x(t) = s(t, \omega t) + f(t, \Omega t). \tag{3.1}$$

Here f is a periodic function and has a zero mean over a complete period as

$$\langle f\rangle = \frac{1}{2\pi}\int_0^{2\pi} f\, d\tau = 0 \tag{3.2}$$

where $\tau = \Omega t$ is the dimensionless time corresponding to the fast time scale. Substituting equations Eqs. (3.1) and (3.2) into Eq. (2.2) we can write the dynamical equation for the slow variable as

$$\ddot{s} + \gamma(s^2 + 2s\langle f\rangle + \langle f^2\rangle - 1)\dot{s} - F_0(t)s + \alpha F_1(s, f) = c\cos\omega t \tag{3.3}$$

and for the fast variable as

$$\ddot{f} + \gamma(s^2 + 2s\,[f - \langle f\rangle] + [f^2 - \langle f^2\rangle] - 1)\dot{f} - F_0(t)f + \alpha F_2(s, f) = g\cos\Omega t \tag{3.4}$$

where the term containing the parametric excitation is given by

$$F_0(t) = \omega_0^2(1 + h\cos\omega_p t). \tag{3.5}$$

The two functions associated with the parameter α in Eqs. (3.3) and (3.4) are given by

$$F_1(s, f) = s^3 + 3s^2\langle f\rangle + 3s\langle f^2\rangle + \langle f^3\rangle \tag{3.6}$$

$$F_2(s, f) = 3s^2(f - \langle f\rangle) + 3s(f^2 - \langle f^2\rangle) + (f^3 - \langle f^3\rangle). \tag{3.7}$$

Now f being a fast variable, one can assume that $\ddot{f}, \dot{f} \gg f, f^2, f^3$ etc, and by invoking this assumption in Eq. (3.4) and then solving it in a self-consistent way [15], by taking only the linear terms, we can arrive at the equation

$$\ddot{f} - \gamma\dot{f} = gB\cos(\Omega t + \eta) \tag{3.8}$$

The amplitude factor of Eq. (3.8) has the form

$$B = \sqrt{(1 + A\cos\beta)^2 + (A\sin\beta)^2} \tag{3.9}$$

where the dimensionless amplitude factor reads

$$A = \frac{\omega_0^2}{\Omega\sqrt{\Omega^2 + \gamma^2}}, \tag{3.10}$$

along with this, the newly introduced phase factors

$$\beta = -\tan^{-1}\left(\frac{\gamma}{\Omega}\right) \tag{3.11}$$

$$\eta = \tan^{-1}\left(\frac{A\sin\beta}{1 + A\cos\beta}\right). \tag{3.12}$$

Eq. (3.8) yields the solution

$$f(t) = \frac{g}{\kappa}\cos(\Omega t + \eta + \theta). \tag{3.13}$$

the associated phase term

$$\theta = -\tan^{-1}\left(\frac{\gamma}{\Omega}\right) \tag{3.14}$$

and the constant κ appearing in the amplitude factor of Eq. (3.13) reads

$$\kappa = \frac{\Omega^2(\Omega^2 + \gamma^2)}{\sqrt{(\Omega^2 - \omega_0^2)^2 + \Omega^2\gamma^2}} \tag{3.15}$$

It can be seen that these constants are proportional to Ω^2 for large values of Ω, so that the contribution of the fast variable $f(t)$ to the main governing dynamics of the system (given by Eq. (2.2)) becomes small, thus justifying the efficacy of this method of solving the differential equation self consistently. Further, the harmonic nature of the solution for $f(t)$ yields the averages

$$\langle f\rangle = \langle f^3\rangle = 0$$

$$\langle f^2\rangle = \frac{g^2}{2}M \tag{3.16}$$

where

$$M = \frac{1}{\kappa^2}. \tag{3.17}$$

Using these above averages along with Eqs. (3.6) and (3.7) in the equation of motion given by Eq. (3.3), we get an effective equation for the slow variable as

$$\ddot{s} + \gamma(s^2 + K)\dot{s} + (\tilde{\omega}^2 - \omega_0^2 h\cos\omega_p t)s + \alpha s^3 = c\cos\omega t \tag{3.18}$$

where the frequency term $\tilde{\omega}$ is given by

$$\tilde{\omega}^2(g) = 3\alpha\left\langle f^2\right\rangle - \omega_0^2 = \frac{3}{2}\alpha g^2 M - \omega_0^2 \tag{3.19}$$

and the factor associated with the nonlinear damping K takes the form

$$K(g) = \frac{g^2}{2}M - 1. \tag{3.20}$$

It is now evident that Eq. (3.18) represents a monostable van der Pol–Duffing oscillator under the influence of the potential

$$V_{mono}(s, t) = \frac{1}{2}\omega_{eff}^2(t)s^2 + \frac{1}{4}s^4 \tag{3.21}$$

where the centrally important effective frequency has the form

$$\omega_{eff} = \sqrt{(\tilde{\omega}^2 - \omega_0^2 h\cos\omega_p t)}. \tag{3.22}$$

A close observation of Eq. (3.18) reveals that a structural modification has been achieved for a specific range of certain parameters. The

coefficient of x in Eq. (2.2) will be negative if we set the parametric excitation strength h to lie within the range $-1 < h < 1$, thus making the oscillator of Eq. (2.2) bistable. This bistability gets modified to an effective monostability if, in addition to the above condition on h, we also set $-\tilde{\omega}^2 < \omega_0^2 h < \tilde{\omega}^2$. This is the central role played by the fast forcing term $g\cos\Omega t$, which in effect changes the natural frequency ω_0 to the new value $\tilde{\omega}$.

4. Slow flow and the resonance response

We rewrite Eq. (3.18) by setting the parametric excitation frequency $\omega_p = 2\omega$

$$\ddot{s} + \gamma(s^2 + K)\dot{s} + (\tilde{\omega}^2 - \omega_0^2 h\cos 2\omega t)s + \alpha s^3 = c\cos\omega t \tag{4.1}$$

Our main objective in this study is to find out the response of the VMD oscillator to the external slow drive as we vary the strength of the rapidly oscillating field. To this end, we specifically focus our attention on the primary resonance, i.e., the response of the system whenever the frequency of the slow drive (ω) is tuned *perturbatively* close to the natural frequency of the slow motion ($\tilde{\omega}$). Rearranging Eq. (4.1) by the use of a dimensionless time $\tau = \omega t$ and defining $\omega = \tilde{\omega} + \epsilon\tilde{\sigma}$

$$s'' + s = \epsilon[-\Gamma(s^2 + K)s' + H(\cos 2\tau)s - \Lambda s^3 + C\cos\tau + \sigma s] \tag{4.2}$$

where we have introduced the detuning parameter σ and the primes denote derivatives with respect to τ. The perturbation parameter ϵ is introduced here to elucidate that all the terms excepting the two on the left-hand side of Eq. (4.2) are small. To this effect, we define the newly introduced parameters on the right hand side of (4.2) as: $\Gamma = \frac{\gamma}{\omega}$, $\Lambda = \frac{\alpha}{\omega^2}$, $C = \frac{c}{\omega^2}$ and $H = \frac{\omega_0^2 h}{\omega^2}$. Also, the σ appearing at the end of the above equation is defined as $\sigma = \frac{2\tilde{\sigma}\tilde{\omega}}{\omega^2}$ and is related to the ratio of the frequencies to first order in ϵ by

$$\frac{\tilde{\omega}^2}{\omega^2} = 1 - \epsilon\sigma \tag{4.3}$$

To derive the amplitude and phase flow equations we apply the multiple time scale analysis [37] to Eq. (4.2) by using perturbative expansion to first order in ϵ,

$$s(\tau_0, \tau_1) = s_0(\tau_0, \tau_1) + \epsilon s_1(\tau_0, \tau_1) + \mathcal{O}(\epsilon^2) \tag{4.4}$$

Here $\tau_i = \epsilon^i\tau$, so that $\tau_0 = \tau$ and $\tau_1 = \epsilon\tau$. The derivatives in terms of τ_i accordingly evaluates to $\frac{d}{d\tau} = \frac{\partial}{\partial\tau_0} + \epsilon\frac{\partial}{\partial\tau_1} + \mathcal{O}(\epsilon^2)$ and $\frac{d^2}{d\tau^2} = \frac{\partial^2}{\partial\tau_0^2} + 2\epsilon\frac{\partial^2}{\partial\tau_1\partial\tau_0} + \mathcal{O}(\epsilon^2)$. Substituting Eq. (4.4) into Eq. (4.2) and equating the coefficient of ϵ order by order we get

$$(\epsilon^0): \quad s_0'' + s_0 = 0, \tag{4.5}$$

$$(\epsilon^1): \quad s_1'' + s_1 = -2\frac{\partial^2 s_0}{\partial\tau_1\partial\tau_0} - \Gamma(s_0^2 + K)s_0' + H\cos(2\tau_0)s_0 - \Lambda s_0^3 + C\cos\tau_0 + \sigma s_0. \tag{4.6}$$

Solving the zeroth-order equation gives

$$s_0(\tau_0, \tau_1) = a(\tau_1)\cos(\tau_0 + \theta(\tau_1)). \tag{4.7}$$

Substituting Eq. (4.7) into Eq. (4.6) and equating the coefficient of secular terms to zero gives the slow flow amplitude and phase equations as follows:

$$\frac{da}{d\tau} = -\frac{1}{2}\left[\frac{\Gamma a^3}{4} + \Gamma K a + C\sin\theta + \frac{Ha}{2}\sin(2\theta)\right] \tag{4.8}$$

$$\frac{d\theta}{d\tau} = \frac{1}{2}\left[\frac{3\Lambda}{4}a^2 - \sigma - \frac{C}{a}\cos\theta - \frac{H}{2}\cos(2\theta)\right] \tag{4.9}$$

The fixed point(s) are obtained by solving the equations (a_0, θ_0) give $\frac{da}{d\tau} = \frac{d\theta}{d\tau} = 0$ which, on combining, give the quadratic equation in σ

$$(\frac{3\Lambda}{4}a_0^2 - \sigma)^2 + (\frac{\Gamma a_0^2}{4} + \Gamma K)^2 = \frac{C^2}{a_0^2} + \frac{H^2}{4} + \frac{CH}{a_0}\cos\theta_0. \tag{4.10}$$

Solving for the detuning parameter we get,

$$\sigma = \frac{3\Lambda}{4}a_0^2 + \sqrt{\frac{C^2}{a_0^2} + \frac{H^2}{4} + \frac{CH}{a_0}Y} \tag{4.11}$$

where the constant Y is the root of a quadratic equation in $\cos\theta_0$ obtained by putting the left-hand side of (4.9) to zero. Thus,

$$Y = \cos\theta_0 = \frac{-\frac{C}{a_0} \pm \sqrt{\frac{C^2}{a_0^2} + 4H\left(\frac{H}{2} + \frac{3\Lambda}{4}a_0^2 - \sigma\right)}}{2H} \tag{4.12}$$

Finally putting back the solution for σ in Eq. (4.11) into Eq. (4.3) and combining that with Eq. (3.19) we get the desired analytical expression for the fast excitation strength g as a function of amplitude a_0,

$$g(a_0) = \sqrt{\frac{1}{M}\left[\frac{2}{3\alpha}\left\{(\omega^2 + \omega_0^2) \pm \epsilon N\right\} - \frac{\epsilon a_0^2}{2}\right]} \tag{4.13}$$

where $N = \sqrt{\frac{c^2}{a_0^2} + \frac{\omega_0^4 h}{4} + \frac{c^2\omega_0^2 h}{a_0}Y}$

This is our sought after response for vibrational resonance. Conventionally, vibrational response is observed by calibrating the high frequency drive strength g. M. Zhan et al. have showed that a simple bistable system can have a enhanced vibrational frequency resonance if one suitably controls the fast signal frequency Ω for a fixed value of g [38]. Following Eq. (4.13) it can be concluded that,our proposed VMD system can also achieve this kind of frequency resonance response for suitably chosen values of parameters. The functional relation between the response amplitude and fast frequency Ω can be visualized as,

$$a_0(\Omega) = \sqrt{\frac{1}{\epsilon}\left[\frac{4}{3\alpha}\left\{(\omega^2 + \omega_0^2) \pm \epsilon N\right\} - 2g^2 M\right]} \tag{4.14}$$

To establish our analytical results we run numerical simulation on Eq. (2.2) as proposed in [1]. The main idea is to extract the Fourier sine and cosine components of the harmonic frequency ω from the mixture of several harmonics present in the nonlinear output signal [x(t)]. We can express the sine and cosine components as

$$Q_s(\omega) = \frac{2}{mT}\int_0^{mT} x(t)\sin(\omega t)dt \tag{4.15}$$

$$Q_c(\omega) = \frac{2}{mT}\int_0^{mT} x(t)\cos(\omega t)dt \tag{4.16}$$

Now, using the RK-4 method to solve $x(t)$ from Eq. (2.2) and then putting back the values into the above equations and evaluating the integrals numerically, we finally get the nonlinear response amplitude

$$a_0(\omega) = \frac{\sqrt{Q_s^2(\omega) + Q_c^2(\omega)}}{c} \tag{4.17}$$

which are shown in (Fig. 1) and (Fig. 2) respectively. Though our analytical expressions of Eqs. (4.13) and (4.14) agree reasonably with the numerical results. However, disagreement gradually creeps in for higher values of Ω; the possible reasons can be twofold. First, the original dynamics of the system is divided into two time scale, a fast and a slow one, from where the rapidly varying variable is averaged out to get an effective slow dynamics of the original system. The numerical integration, on the other hand, preserves all the time scales. Second, to derive the flow equations (Eqs. (4.8) and (4.9)) we restrict our calculations to order $\mathcal{O}(\epsilon)$ as opposed to the numerical approach that is ideally capable of reproducing almost exact results. This point, however, is generically applicable for all perturbative approaches.

Finally (Fig. 3) states the variation of g_{max}(i.e. the value of fast drive strength g at response peak) with Ω and c.

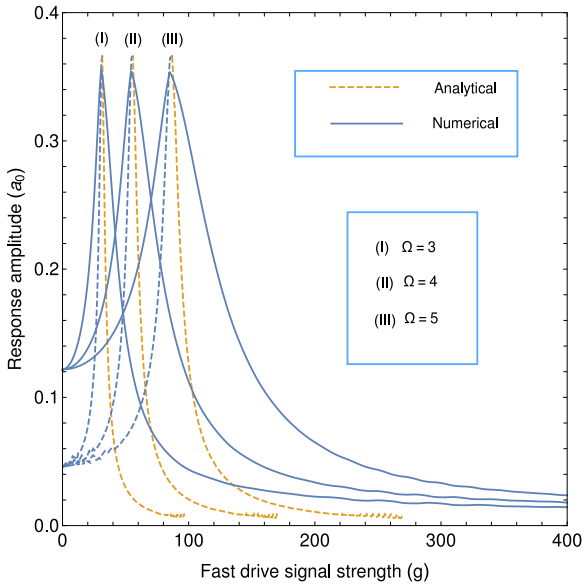


Fig. 1. Response amplitude (a_0) as function of the strength (g) of the high-frequency excitation for a fixed set of parameters $\omega_0 = 0.3$, $h = 0.01$, $\gamma = 0.15$, $\alpha = 0.01$, $c = 0.05$ and $\omega_p = 0.6\ \omega = 0.3$, at different values of Ω. Analytical plots (dashed lines) are based on Eq. (4.13) and the numerical results (solid lines) are based on Eq. (2.2).

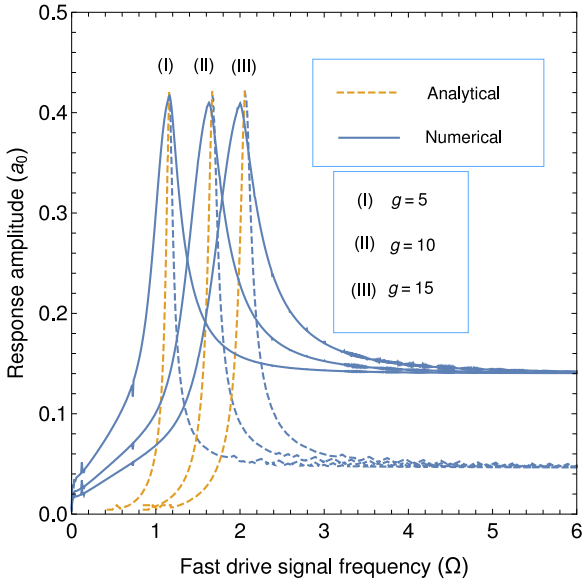


Fig. 2. Response amplitude (a_0) as function of frequency Ω of the high-frequency excitation for a fixed set of parameters $\omega_0 = 0.3$, $h = 0.01$, $\gamma = 0.15$, $\alpha = 0.01$, $c = 0.05$ and $\omega_p = 0.6\ \omega = 0.3$, at different values of g. Analytical plots (dashed lines) are based on Eq. (4.14) and the numerical results (solid lines) are based on Eq. (2.2).

5. Limit cycle and hopf bifurcation of the slow dynamics

The flow equations (Eqs. (4.8) and (4.9)) obtained for the slow variable by averaging out the fast variable from Eq. (2.2) can be further studied with the purpose of exploring the behaviour of limit cycle which is characteristic of a van der Pol oscillator. To understand the behaviour of the flow equations, we have to resort to perturbation theory yet again by introducing a new perturbation parameter λ. The limit cycle that transpires from the subsequent analysis represents an oscillation of the slow variable over a much longer time scale, viz., $t \sim \mathcal{O}(1/\lambda\epsilon)$. In the literature, the new set of flow equations thus obtained by effecting a perturbative expansion for a second time goes by the

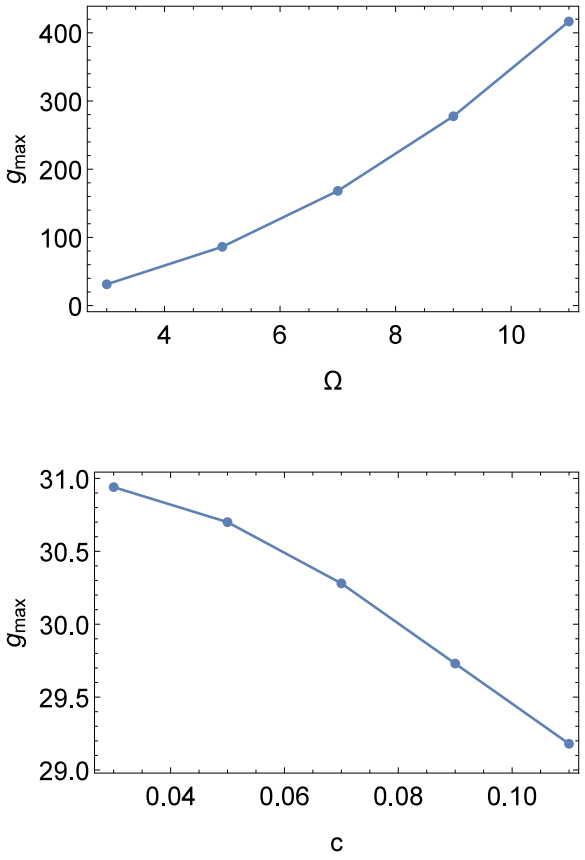


Fig. 3. Numerical result for the position of maximum g_{max} of the nonlinear response amplitude: (a) as a function of fast frequency (Ω) and for $c = 0.05$, $\gamma = 0.15$, $\alpha = 0.01$ and $\omega = \omega_p = 0.3$. (b) as a function of the slow-frequency signal strength c for $\Omega = 3$, $\gamma = 0.15$, $\alpha = 0.01$ and $\omega = \omega_p = 0.3$.

name "super-slow flow" equations [39]. These equations eventually allow us to predict, with significant accuracy, the value of the control parameter g at which the effective slow dynamics goes through a *Hopf bifurcation* thus leading to the birth of a stable limit cycle.

5.1. The super-slow flow equations

To derive the super-slow flow equations, we parametrize the slow flow equations (Eqs. (4.8) and (4.9)) as

$$u = a\cos\theta \qquad \text{and} \qquad v = -a\sin\theta \tag{5.1}$$

so that,

$$\frac{du}{d\tau} = (-\frac{\sigma}{2} + \frac{H}{4})v + \lambda\Big\{\frac{1}{8}(3\Lambda v - \Gamma u)(u^2+v^2) - \frac{\Gamma K u}{2}\Big\} \tag{5.2}$$

$$\frac{dv}{d\tau} = (\frac{\sigma}{2} + \frac{H}{4})u + \lambda\Big\{\frac{c}{2} - \frac{1}{8}(3\Lambda u + \Gamma v)(u^2+v^2) - \frac{\Gamma K v}{2}\Big\} \tag{5.3}$$

where the newly brought in perturbation parameter λ has been invoked in such a way that the zeroth-order terms yield simple harmonic solutions for the variables u and v. This arrangement requires that we cluster the (modified) damping constant (Γ), nonlinearity (Λ) and the strength of the slow drive (c) under the common banner of the new perturbation parameter λ. In the spirit of multiple time scale analysis,

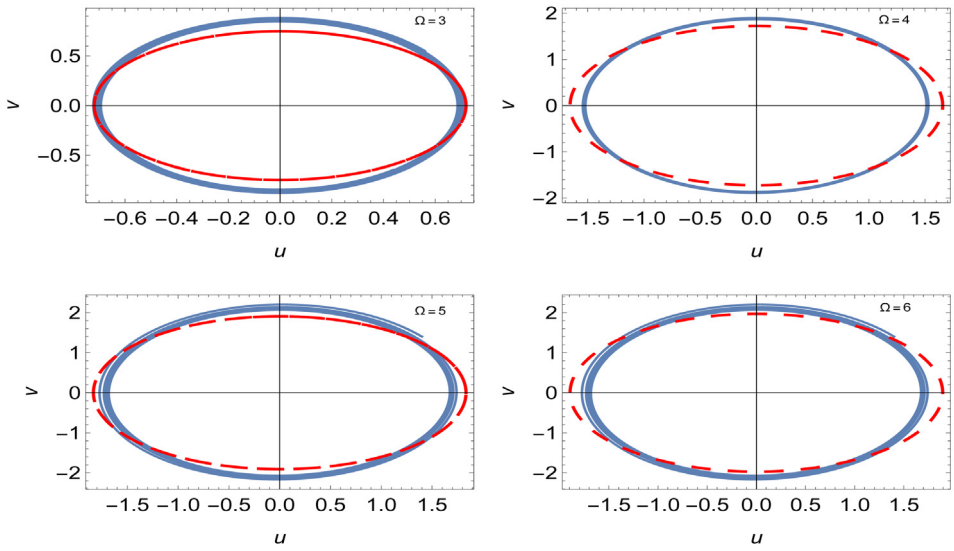


Fig. 4. Numerical(solid lines) and analytical(dashed line) comparison of limit cycles for different values of $\Omega's$ and for fixed value of fast frequency drive strength g. Simulation is done on Eqs. (5.2) and (5.3) where the analytical plot based on Eqs. (5.9) and (5.10). And the related parameters $\gamma = 0.0015, \alpha = 0.0001, c = 0.0005, \omega = \omega_0 = 0.3, h = 0.01$ and $g = 12$.

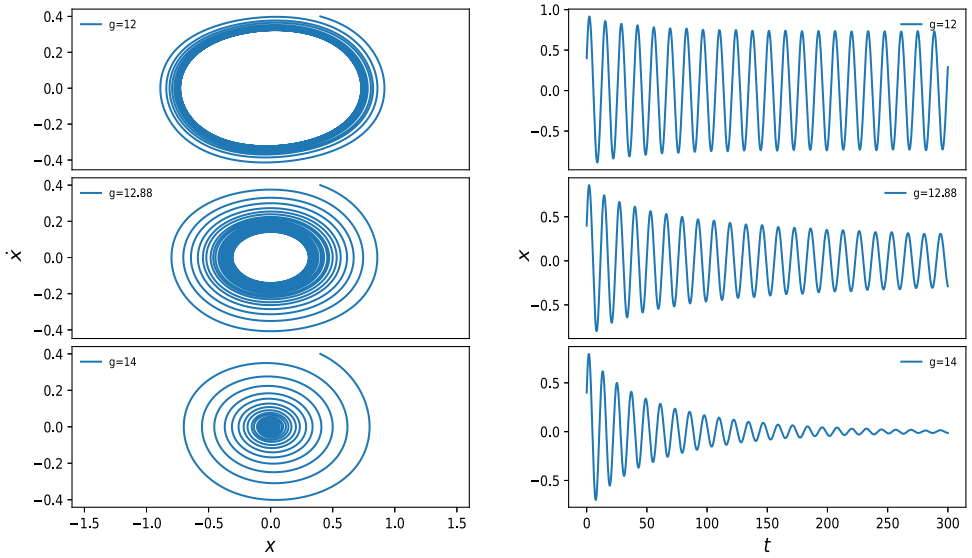


Fig. 5. Numerical simulation of Supercritical Hopf bifurcation based on Eq. (4.1) for $\Omega = 3$. Other parameters are fixed and given by $\gamma = 0.15, \alpha = 0.1, c = 0.0005, \omega = \omega_0 = 0.3$ and $h = 0.01$. Shows a clear transition from stable limit cycle to stable spiral at $g_{hopf} = 12.88$.

we now invoke the perturbation expansions [40,41] in λ as,

$$u(T_0, T_1) = u_0(T_0, T_1) + \lambda u_1(T_0, T_1) + \mathcal{O}(\lambda^2)$$
$$v(T_0, T_1) = v_0(T_0, T_1) + \lambda v_1(T_0, T_1) + \mathcal{O}(\lambda^2) \tag{5.4}$$

where $T_j = \lambda^j \tau$. Now the derivatives in terms of variables T_i are connected to τ as $\frac{d}{d\tau} = D_0 + \lambda D_1 + \mathcal{O}(\lambda^2)$ where $D_j = \frac{\partial}{\partial T_j}$. Substituting the set of equations Eq. (5.4) into Eqs. (5.2) and (5.3) followed by separation of the terms in accordance with the powers of λ we get, the zeroth-order and the first order equations as follows:

$$(\lambda^0): \quad D_0^2 u_0 + \omega_l^2 u_0 = 0 \tag{5.5}$$
$$-(\frac{\sigma}{2} - \frac{H}{4}) v_0 = D_0 u_0 \tag{5.6}$$

$$(\lambda^1): \quad D_0^2 u_1 + \omega_l^2 u_1 = -(\frac{\sigma}{2} - \frac{H}{4})\Big[-D_1 v_0 + \frac{c}{2} - \frac{\Gamma K v_0}{2} - \frac{1}{8}(3\Lambda u_0 + \Gamma v_0)(u_0^2 + v_0^2)\Big] - D_0 D_1 u_0 - \frac{\Gamma K}{2} D_0 u_0 + D_0\Big[\frac{1}{8}(3\Lambda v_0 - \Gamma u_0)(u_0^2 + v_0^2)\Big] \tag{5.7}$$

$$-(\frac{\sigma}{2} - \frac{H}{4}) v_1 = D_0 u_1 + D_1 u_0 + \frac{\Gamma K u_0}{2} - \frac{1}{8}(3\Lambda v_0 - \Gamma u_0)(u_0^2 + v_0^2) \tag{5.8}$$

where the super-slow oscillation frequency is defined as $\omega_l = \sqrt{\frac{\sigma^2}{4} - \frac{H^2}{16}}$. From Eqs. (5.5) and (5.6) we obtain the zeroth-order solution as

$$u_0 = r(T_1)\cos(\omega_l T_0 + \psi(T_1)) \tag{5.9}$$
$$v_0 = \frac{\omega_l}{p} r(T_1)\sin(\omega_l T_0 + \psi(T_1)) \tag{5.10}$$

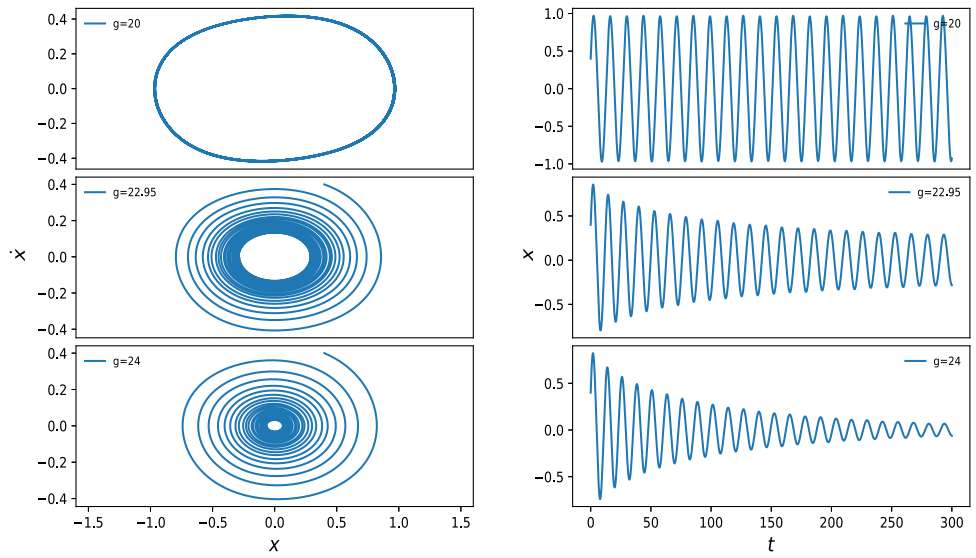


Fig. 6. Numerical simulation of Supercritical Hopf bifurcation based on Eq. (4.1) for $\Omega = 4$. Other parameters are fixed and given by $\gamma = 0.15, \alpha = 0.1, c = 0.0005, \omega = \omega_0 = 0.3$ and $h = 0.01$. Shows a clear transition from stable limit cycle to stable spiral at $g_{hopf} = 22.95$.

where we define $p = -(\frac{\sigma}{2} - \frac{H}{4})$. Now invoking Eqs. (5.9) and (5.10) into Eq. (5.7) and equating the secular terms to zero generates new set of flow equations as

$$D_1 r = \frac{-\Gamma K}{2} r - \frac{\sigma \Gamma}{8\sigma - 4H} r^3 \tag{5.11}$$

$$D_1 \psi = -\frac{9\Lambda\sigma^2}{4(2\sigma - H)\sqrt{(4\sigma^2 - H^2)}} r^2. \tag{5.12}$$

The equilibrium solution $D_1 r = 0$ gives the analytical estimate of the radius of the limit cycle

$$r = 0 \quad \text{and} \quad r = \sqrt{K(\frac{2H}{\sigma} - 4)} \tag{5.13}$$

where, through a linear stability analysis, we see that the origin is unstable while the second root of r, denoting the limit cycle, is stable. Clearly, this non-trivial root of r reveals the dependence of the amplitude of the limit cycle on the fast drive strength g and the frequency Ω through $K(g)$ defined in Eq. (3.20). This analytical expression complies reasonably with the results obtained by numerically solving the system given by Eqs. (5.2) and (5.3) for different sets of (Ω), as has been shown in (Fig. 4).

From Eq. (5.12), it is also clear that in the parametric space, $K = 0$ represents a point where a supercritical Hopf bifurcation occurs. When $K > 0$, the origin acts as the only stable fixed point. Looking back at Eq. (3.20) we see that by changing the control parameter g, we can change K. Therefore, when K becomes negative, we see that the origin is destabilized, and a stable limit cycle is created. This transition occurs at the point $K = 0$ which, according to Eq. (3.20), happens at the threshold value of g given by

$$g_{hopf} = \sqrt{\frac{2}{M}}. \tag{5.14}$$

With the values of the parameters chosen as $\gamma = 0.15$, $\alpha = 0.1$, $c = 0.0005$, $\omega = \omega_0 = 0.3$ and $h = 0.01$, combining Eqs. (3.15) and (3.17) we get $M = 0.012$ for $\Omega = 3$ and $M = 0.0038$ for $\Omega = 4$. Placing them one by one into Eq. (5.14) gives $g_{hopf} = 12.88$ and $g_{hopf} = 22.95$ respectively for $\Omega = 3$ and $\Omega = 4$. These values are in very good agreement with the data obtained by numerically solving the equation of motion of the slow variable given by Eq. (4.1), as have been shown in Figs. 5 and 6.

6. Conclusions

To summarize, we have studied the parametric resonance of a van der Pol–Duffing oscillator under the influence of two periodic drives with widely different frequencies and have shown that the response of the system to the slow drive can be controlled by modulating the strength of the fast drive, g. Studies pertaining to *vibrational resonance* are usually modelled in this way. Although in literature there have been some contributions concerning the study of this parametric VMD oscillator driven by high-frequency excitation [23–25], the study and interpretation of nonlinear response in the backdrop of *vibrational resonance* on this type of system have, to the best of our knowledge, not been prominently pursued. The procedure of sieving out an effective slow dynamics by averaging out the fast variable has two important consequences. First, the symmetrically oscillating bistable potential (with the natural frequency ω_0) figuring in the starting differential equation Eq. (2.2) gets reshaped to a symmetrically oscillating monostable potential with a new frequency $\tilde{\omega}$ within a certain range of values of the parameters involved. In majority of works regarding parametric resonance, dynamics close to the natural frequency (ω_0) or the parametric frequency (ω_p) have been studied, and the primary resonance is usually defined accordingly. In this work, however, we have studied the dynamics by deriving flow equations for amplitude and phase, by effecting perturbation theory in the vicinity of the new frequency ($\tilde{\omega}$), and not around ω_0 or ω_p, thus leading to consequences not explored in earlier works. Second, the strength of the fast drive, g, also reshapes the damping by contributing an additional nonlinearity in the damping term of the initial dynamics described by Eq. (2.2). This modification has a pronounced effect on the formation of limit cycle in the self-excited oscillator in that, by modulating g, one can control the position of the point in parameter-space where the all-important Hopf bifurcation should occur. In this context, we have been able to define a threshold value g named g_{hopf} below which the limit cycle should exist. By keeping g in this range, viz. $g < g_{hopf}$, one can adjust the fast frequency Ω as well as the strength of the parametric oscillation h so that there is a stable limit cycle oscillation in an effective monostable potential with a frequency given by ω_l evaluated in Section 5.

The present study may have important applications in nanomechanical oscillatory systems [42,43]. Extensions of this study to understand the subharmonic and superharmonic responses for parametrically driven oscillators in the backdrop of vibrational resonance can have many interesting implications.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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